NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-6106/41A(USAF)
27 December 1989

SUPERSEDING
MIL-R-6106/41(USAF)
11 February 1983

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, 25 AMPERES, 3PNO, WITH 2 AMPERE 1PDT AUXILIARY CONTACTS, HERMETICALLY SEALED, TYPE I

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

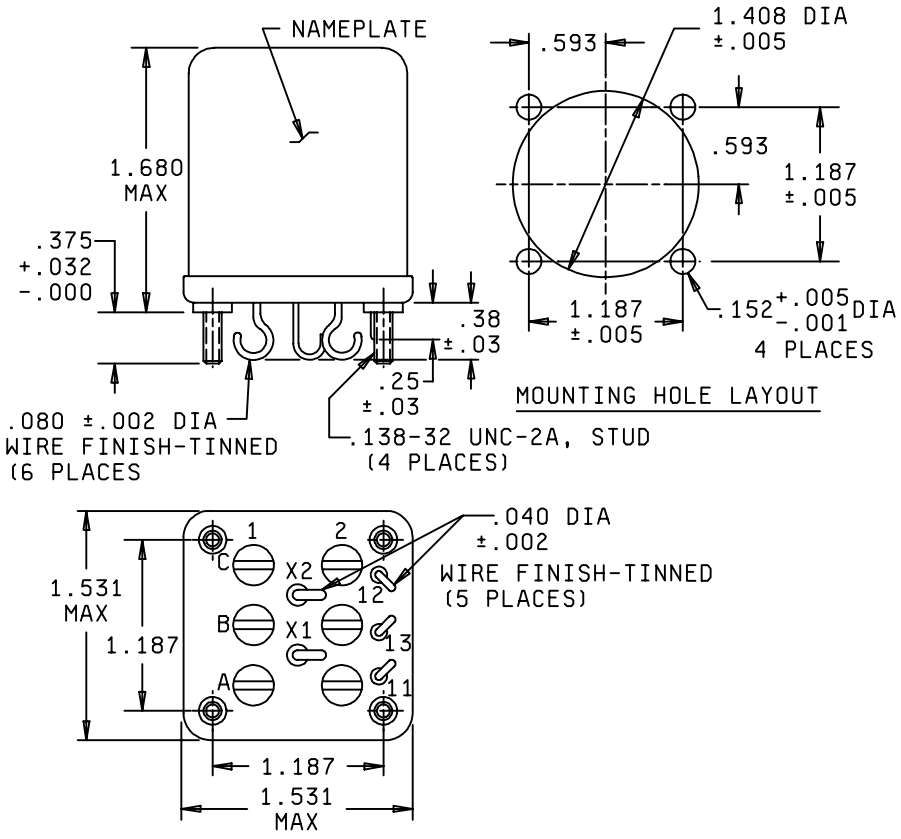
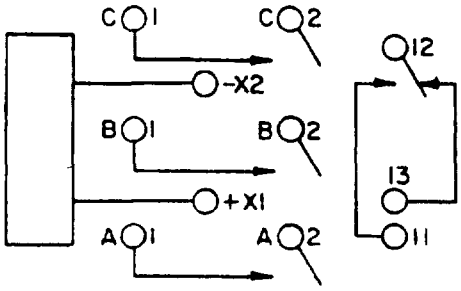


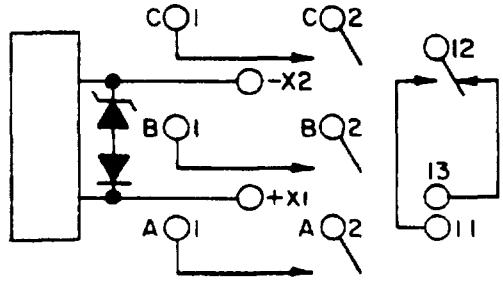
FIGURE 1. Dimensions and configurations.

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Ⓐ Circuit diagram
(dash numbers -001, -002, -003)
(see note 5)



Ⓐ Circuit diagram
(dash number -004)

Inches	mm	Inches	mm
.001	0.03	.25	6.4
.002	0 05	.375	9.53
.005	0.13	.38	9.7
.03	0.8	.593	15.06
.032	0.81	1.187	30.15
.040	1.02	1.406	35.71
.080	2.03	1.531	38.89
.152	3.86	1.680	42.67

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is .010 (0.25 mm).
4. Terminal numbers need not appear on the relay header provided there is affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.

FIGURE 1. Relay, outline drawing - Continued.

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REQUIREMENTS:

Coil data.

Coil data: See table I.

Duty rating: Continuous.

Operational data:

Rated contact load. See table II.

Operate time: See table I.

Release time: See table I.

(A) Maximum induced transient voltage (when specified): 42 V.

Physical.

Dimensions and configurations. See figure 1.

Weight (maximum): 0.44 pound (200 grams).

Terminal type: See figure 1 and table I.

Finish: Flat black.

ENVIRONMENTAL CHARACTERISTICS:

Temperature rating: -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

Shock, g-level: 50 g.

Duration: 11 ms.

Maximum duration contact opening: 2 ms.

Vibration - sinusoidal:

Operating:

G-level: 10 g.

Frequency range. 5 to 1,500 Hz.

Nonoperating:

G-level: 15 g.

Frequency range: 20 to 2,000 Hz.

Acceleration: 10 g.

ELECTRICAL CHARACTERISTICS:

Insulation resistance:

Initial. 100 megohms.

After life or environmental tests: 50 megohms.

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TABLE I. Mounting, termination, and operating characteristics.

	Dash number	Coil terminal X1, X2	Coil data: Nominal Volts 1/	Nominal Freq	Nominal Ⓐ Res ±10%	Maximum Volts	Maximum Amp	Max pickup voltage: Nominal 2/	Max pickup voltage: High temp test	Max pickup voltage: Cont current test	Hold voltage 2/	Drop out voltage 2/	Time (milliseconds max): Operate 3/	Time (milliseconds max): Release 4/	Contact bounce: Main NO	Contact bounce: Main NC	Contact bounce: Aux NO	Contact bounce: Aux NC	Terminals	Mounting
	-001	DC	28	DC	---	29	0.25	18	19.5	22.5	7.0	1.5	20	10	2.0	---	4.0	4.0	Solder-hook	4 Studs
	-002	AC	115	400	---	122	.055	96	100	108	40	5.0	25	50	2.0	---	4.0	4.0	Solder-hook	4 Studs
	-003	AC	115	50/60	---	122	.06	96	100	108	40	5.0	25	50	2.0	---	4.0	4.0	Solder-hook	4 Studs
Ⓐ	-004	DC 5/	28	DC	160	29	0.25	18	19.5	22.5	7.0	1.5	20	10	2.0	---	4.0	4.0	Solder-hook	4 Studs

1/ CAUTION. Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
Ⓐ 5/ Suppressed dc coil: Back EMF 42 volts maximum.

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TABLE II. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate footnote
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	50/60 Hz	400 Hz	60 Hz	400 Hz	50/60 Hz	400 Hz	60 Hz	
Resistive	50	25	---	2	2	25	25	2	---	25	25	---	---	---
Inductive	10	15	---	1	1	25	25	1	1	25	25	---	---	2/
Motor	50	20	---	---	---	20	12	---	---	20	12	---	---	---
Lamp	50	10	---	.5	.5	10	10	.5	---	10	10	---	---	---
Transfer load		---	---	---	---	---	---	---	---	---	---	---	---	3/
Mechanical life reduced current	100	6.3	---	.5	.5	6.3	6.3	.5	---	6 3	6.3	---	---	---
Ⓐ Intermediate current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3 phase applications.
2/ .7 pF, inductive.
3/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Dielectric withstanding voltage (sea level)

	Initial	After life tests
Coil to case	1,250 V rms	1,000 V rms
Auxiliary contacts	1,000 V rms	750 V rms
All other points	1,500 V rms	1,125 V rms

Dielectric withstanding voltage (altitude):

	80,000 feet
Coil to case	500 V rms
Auxiliary contacts	250 V rms
All other points	500 V rms

Maximum contact drop·

Initial: 0.150 volt

After life test: 0.175 volt.

Overload current 80 amperes dc, 120 amperes ac

Rupture current: 100 amperes dc, 150 amperes ac.

RFI specification (applicable to coil currents of ac operated relays)· MIL-STD-461.

PART NUMBER M6106/41 (plus dash number from table I)

CONCLUDING MATERIAL

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent·
DLA - ES

(Project 5945-F701)

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